

Pandas Basics

1. What is Pandas?

Pandas is an open-source Python library used for data manipulation, analysis and cleaning. It provides fast, flexible, and easy-to-use **data structures** for working with **structured data** like tables. Pandas provides two data structures such as Series and DataFrame structures for efficient data manipulation and analysis.

2. What is a DataFrame?

A Pandas DataFrame is a two-dimensional table-like structure in Python where data is arranged in rows and columns. It's one of the most commonly used tools for handling data and makes it easy to organize, analyze and manipulate data. It can store different types of data such as numbers, text and dates across its columns. The main parts of a DataFrame are:

- **Data:** Actual values in the table.
- **Rows:** Labels that identify each row.
- **Columns:** Labels that define each data category.

```
import pandas as pd

lst = ['Geeks', 'For', 'Geeks', 'is',
       'portal', 'for', 'Geeks']

df = pd.DataFrame(lst)
print(df)
```

Output:

	0
0	Geeks
1	For
2	Geeks
3	is
4	portal
5	for
6	Geeks

Output

3. Difference between Series and DataFrame

Series	DataFrame
One-dimensional data structure	Two-dimensional data structure
Series elements must be homogeneous	Can store heterogeneous data
Immutable (size cannot be changed)	Mutable (size can be changed)
Supports element-wise computations	Supports column-wise computations
Provides less functionality	Provides more functionality
Alignment not supported	Alignment supported
Has only index labels	Has both row index and column labels
Represents a single column of data	Represents a table of multiple columns
Used for single variable data	Used for multiple variable datasets

4. What is CSV?

CSV (Comma-Separated Values) is a plain text file format used to store tabular data, where each value is separated by a comma. Data is stored in rows and columns. It allows users to load tabular data into a DataFrame, which is a powerful structure for data manipulation and analysis. To access data from the CSV file, we require a function `read_csv()` from Pandas that retrieves data in the form of the data frame.

5. Why Pandas is important for AI?

- **Data Cleaning and Preparation**
 - Helps sort, clean, and preprocess large volumes of raw data

- Handles missing values, duplicates, and inconsistencies efficiently
- **Exploratory Data Analysis (EDA)**
 - Quickly summarizes datasets using descriptive statistics
 - Helps understand patterns, trends, and relationships in data
- **Feature Engineering**
 - Assists in selecting, transforming, and creating important features
 - Makes data suitable for machine learning models
- **Efficient Data Handling**
 - Supports structured data formats like CSV, Excel, SQL, and JSON
 - Enables fast filtering, slicing, and aggregation
- **Integration with AI/ML Libraries**
 - Works seamlessly with NumPy, Scikit-learn, TensorFlow, and PyTorch
 - Easily converts data into formats required by ML models
- **Time-Series and Trend Analysis**
 - Provides tools for date-time handling and time-series analysis
 - Useful for trend detection and forecasting
- **Future Analysis and Prediction Support**
 - Prepares historical data for predictive modeling
 - Plays a key role in forecasting future trends in AI applications